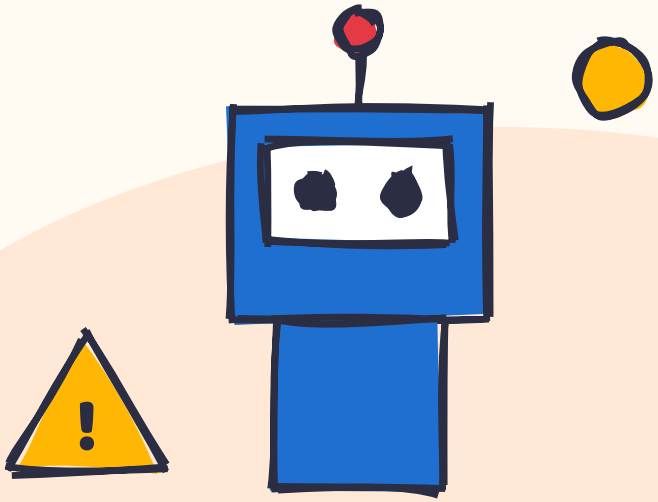
Little Explorers · Class 31

Safety Helper Robot

Invent a robot that keeps people safe!

Danger Design Present



Did you know?
Smoke alarms are little safety robots: they smell smoke and shout BEEP to warn everyone!

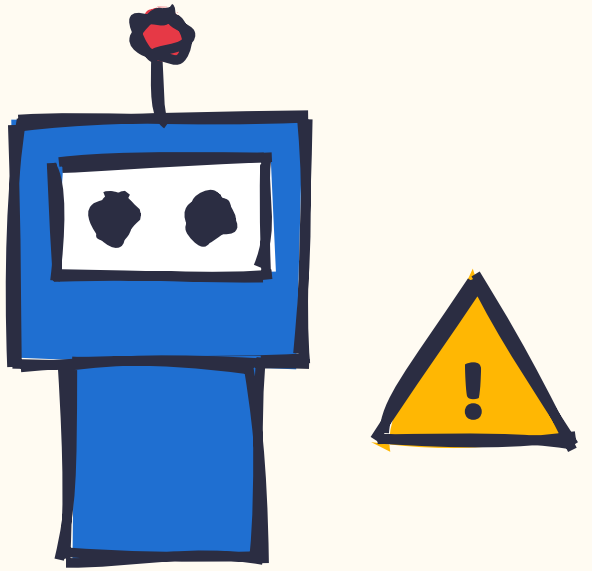
LEARN

Let's wonder...

Who keeps us safe all day?

Watch out!

A wet floor. A hot pan. A busy road outside the school gate. Grown-ups can't watch everything all the time. Could a **robot helper** watch for us?



Teachers do!

A dog?

A robot!

Today: you become **inventors**: you design and model a robot that keeps people safe.

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Learning objectives

Today I can…

1	**Spot the danger:** find a **safety problem** at school or at home	**Proof:** I named a danger a robot could help with
2	**Feel, think, show:** design a robot with **sensors** (feel), a **rule** (think) and an **alarm** (show)	**Proof:** my drawing shows all 3 parts
3	**Present it:** build a **model** and tell the class how my robot keeps people safe	**Proof:** I presented my robot

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Objective 1 · Spot the danger · video

Video time!



Safety Tips for Kids

learning junction · about 5 minutes · youtu.be/CqH2QYt6oOc

Watch for...

1. Name 3 **safety rules** from the video.
2. What should you do if you get lost?
3. Who can you ask for help?

Scan to watch

After the video: Which safety rule could a ROBOT help with?

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Objective 1 · Spot the danger

Spot the danger

The idea

Inventors start with a **problem**, not a robot! Look around: where could someone get **hurt**?

Slippery floors, hot things, busy roads, things left on the stairs, getting lost…

Example: a wet floor sign



wet floor!

When the floor is mopped, a yellow sign warns everyone. It's a simple safety helper. A robot could put the sign out by itself!

Objective 2 · Feel, think, show

Feel, think, show

The idea

Every helper robot has 3 parts:

Feel: a sensor notices something (heat, water, movement).

Think: a rule decides (**if** too hot, **then**...).

Show: it warns us with a light, a sound or a message.

Example: a smoke alarm



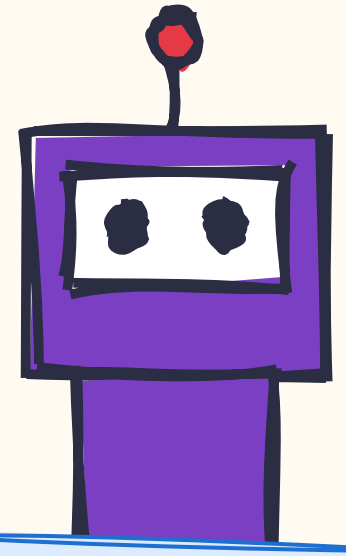
Feel: it smells smoke. Think: if smoke, then danger! Show: BEEP BEEP BEEP. That's feel, think, show!

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Ideas

Helper robot examples

- **Wet-floor bot:** feels water on the floor, then flashes and says 'Slippery!'
- **Hot-pan bot:** feels heat on the stove, then beeps if a child comes near
- **Gate guard:** sees cars at the school gate, then shows a red STOP light
- **Lost-child buddy:** hears 'help!', then calls a teacher



Your own idea

These are just ideas. The best robot fixes a danger YOU noticed!

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Real world

Safety helpers we already have



Smoke alarm
Smells smoke → beeps



Car sensors
Beep when you back up near a wall



Motion light
Lights the path at night



Crossing light
Tells us when to cross



Which safety helper have you seen this week?

Feel, think, show
Every one of these has a sensor, a rule and an alarm!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Danger detectives
2. Design on paper
3. Build a model
4. Practise your pitch
5. Inventor fair score card

Class game

Danger detectives

How to play

1. Teams walk around the classroom (or look at picture cards).
2. Find **3 places** where someone could get hurt.
3. For each one, say: what could a robot **feel**?
4. Share your best danger with the class.

Then discuss

Which danger would be the most useful to fix?



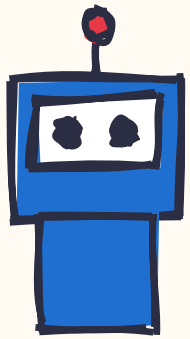
Big idea
Good inventions start with a real problem.

MAKE

Objective 2 · Build step 1 of 3

Design on paper

- **a** Choose your **danger**.
- **b** Draw your robot. Give it a name!
- **c** Draw its **sensor** (feel) and label it.
- **d** Write or draw its **rule**: 'If ___, then ___'.
- **e** Draw how it **warns** people (show).



If hot, then BEEP!

Check: Your drawing has feel, think and show.

MAKE

Objective 3 · Build step 2 of 3

Build a model

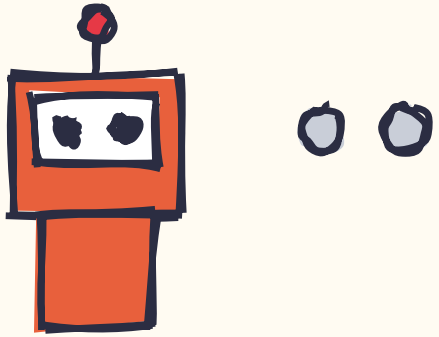
a Use a box for the body and a cup for the head.

b Bottle caps = sensors. Foil = lights. Pipe cleaners = antennae.

c Stick on labels: **FEEL**, **THINK**, **SHOW**.

d Use real LEDs from our kit if you want lights!

e Make it look friendly.



Check: Your model shows where each part is.

Scissors: cut away from your body. Ask for help with thick card.

Objective 3 · Build step 3 of 3

Practise your pitch

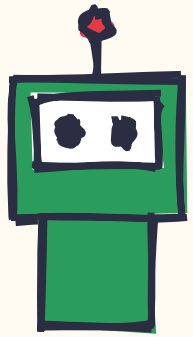
a Practise saying:

b 'My robot is called ___.'

c 'It keeps people safe from ___.'

d 'It feels ___, and if ___, then it ___.'

e Show where the sensor is on your model.



Hello!

Check: You can explain your robot in 4 sentences.

MAKE

Show time!

Inventor fair score card

Robot name	Danger it fixes	Feel, think, show?
Robot 1		
Robot 2		
Robot 3		

Talk about it

Which robot would you most like at our school? Why?

Kind words

Say one thing you like about each robot before asking a question.

Objective check

Can I really do it? Prove it.

01 What danger does your robot help with? ○ ○ ○

02 Point to the feel, think and show parts. ○ ○ ○

03 Tell the class your robot's rule. ○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Inventor fair: every team presents, and the class gives a clap and one kind comment.

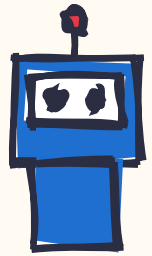
Show what you know

What did we learn?

- [x] Inventors start with a real problem
- [x] Helper robots feel, think and show
- [x] Rules sound like: if ___, then ___
- [x] Explaining your idea is part of inventing

At home

Find one safety helper at home (smoke alarm, light, sign) and tell the class.



Exit ticket: what are the 3 parts of a helper robot?

Next class:
Countdown
Rocket Launch!